

TESTING THE WATERS
Scientists and Sikkim government officials conduct hydrological tests in the Lhonak valley

THE NEW PATH

- In October 2023, the South Lhonak lake glacial flood devastated Sikkim, destroying infrastructure and ecology and displacing thousands
- To develop a model to mitigate Glacial Lake Outburst Flood risks, a preliminary study with NDMA identified 180 high-risk lakes in the region, 40 of which are in Sikkim
- Expedition teams of scientists, NDMA, Sikkim state officials trekked to high-altitude lakes and carried out various tests
- After this, Sikkim devised a four-step approach to glacial flood risk mitigation

Management Authority and Sikkim University. The nine expeditions, lasting about 12 days and costing Rs 15 lakh each, were a first-of-their-kind effort. The remaining seven lakes, inaccessible in winter, are slated for study in the coming summer. Insights from the completed assessments have informed mitigation strategies. Real-time monitoring systems, lake water level controls, retention structures at the valley level, flood walls, enhanced building codes, public awareness, and insurance initiatives form the cornerstone of these plans. "The interdisciplinary studies would help develop site-specific technical proposals to safeguard the state from future GLOFs. Specially posting engineers to the department provided expertise in designing these structural measures," says Pintso Namgyal Lepcha, minister of the Science and Technology department.

"Sikkim has devised a four-step approach to glacial flood risk mitigation. By setting up this standard operating procedure, it is pioneering a method," says Sandeep Tambe, secretary of the department and a veteran of two expeditions. According to Dr Gourab Bhattacharya, assistant professor in the Department of Earth Sciences in the Tennessee Tech University, "mapping" is the first step. "It's necessary to have automated early warning systems so that evacuation protocols can be initiated downstream when glacial dams are breached," he says. He prescribes routine inspection of hydroelectric plants downstream of the Himalayan rivers by geotechnical engineers.

As Sikkim prepares for its next set of expeditions, the journey ahead remains fraught with challenges. Yet, the state's innovative approach underscores a broader message: adaptation and preparedness can turn adversity into opportunity. By safeguarding its people and environment, Sikkim is not just recovering from the devastation, but declaring itself to be the avant garde in climate resilience. ■